

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	G.Hima Bindu
Register Number	192511377

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Hima Bindu*

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

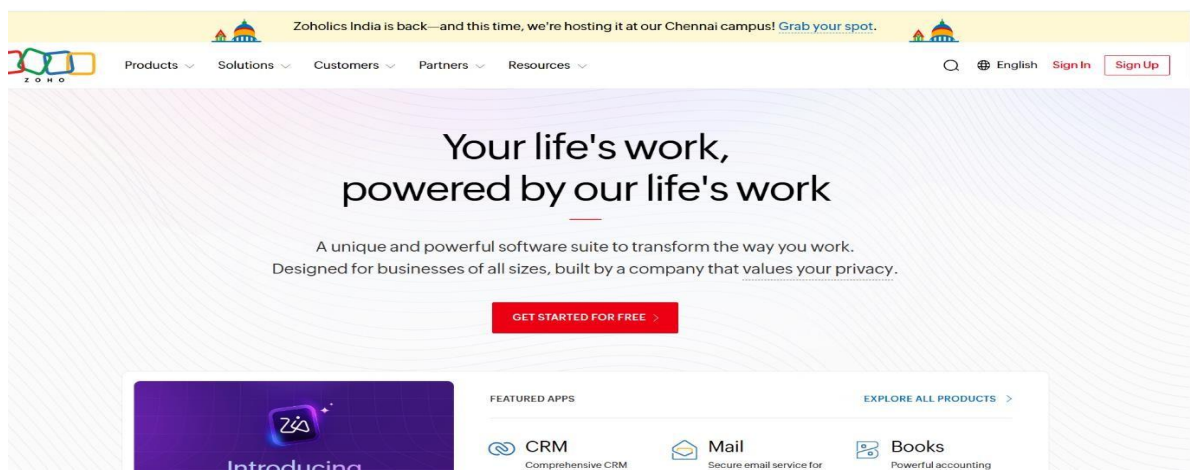
To develop a simple cloud-based Library Book Reservation System for SIMATS Library using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing book details, availability, and reservations.

ALGORITHM:

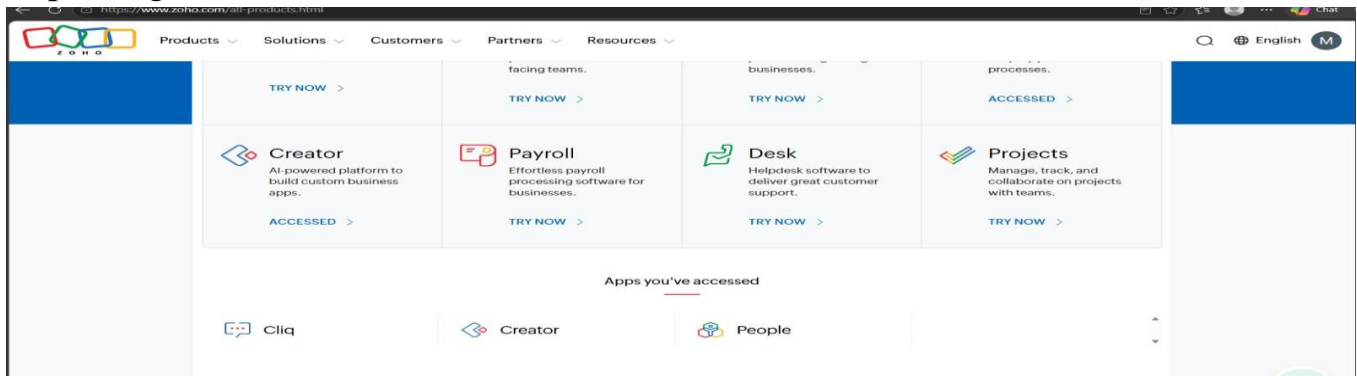
1. Start
2. Open the Library Book Reservation System.
3. Enter the book details such as Title, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.
4. Store the details in the cloud database.
5. Search for the required book.
6. Check whether the book is Available/Taken.
7. If available, reserve the book and update the status as Taken.
8. If the book is returned, update the status as Returned/Available.
9. Display the updated book details.
10. Stop.

Implemented:

Step1: Go to Zoho .com

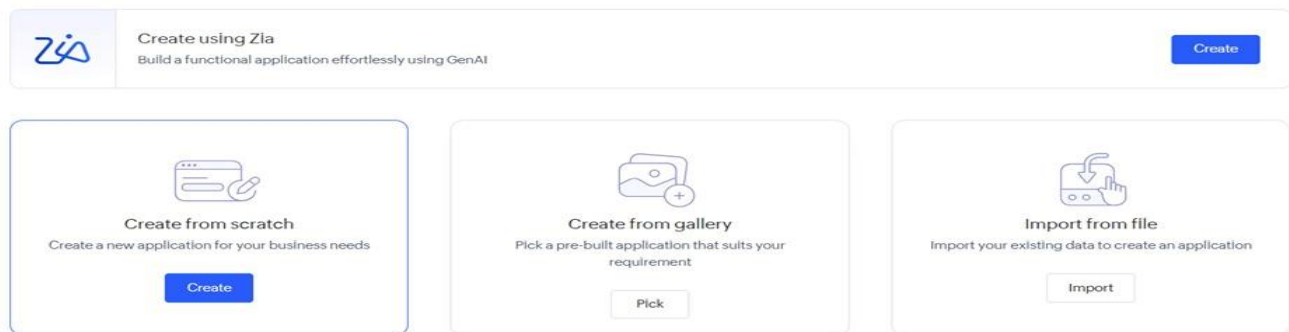


Step2: Login to Zoho.Com



Step3: Create New Application from Select from Scratch

Create New Application



STEP 4: Enter Library Book Reservation System as the application name and click Create Application.

A screenshot of the 'Create from scratch' form. At the top, there is a logo consisting of four circles (two blue, two white) and a plus sign. Below the logo, the text 'Create from scratch' is displayed. The form contains a label 'Application Name' with a red asterisk, followed by a text input field containing 'Library Book Reservation System'. Below the input field, there is a checkbox labeled 'Enable Environment' with a link 'Learn More'. At the bottom of the form, there is a large blue button labeled 'Create Application'.

Step 5:Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

Library Book Reservation

Dashboard

Book Copies

User Profiles

Reservations

+ Reservations

Reservations Report

Board

Books

Users

Himabindu G...

Reservations

Book Copy

User

Reservation Date

Due Date

Return Date

Status

Created Time

Last Modified Time

Submit

Reset

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

Library Book Reservation

Dashboard

Book Copies

User Profiles

Reservations

+ Reservations

Reservations Report

Board

Books

Users

Himabindu G...

Reservations Report

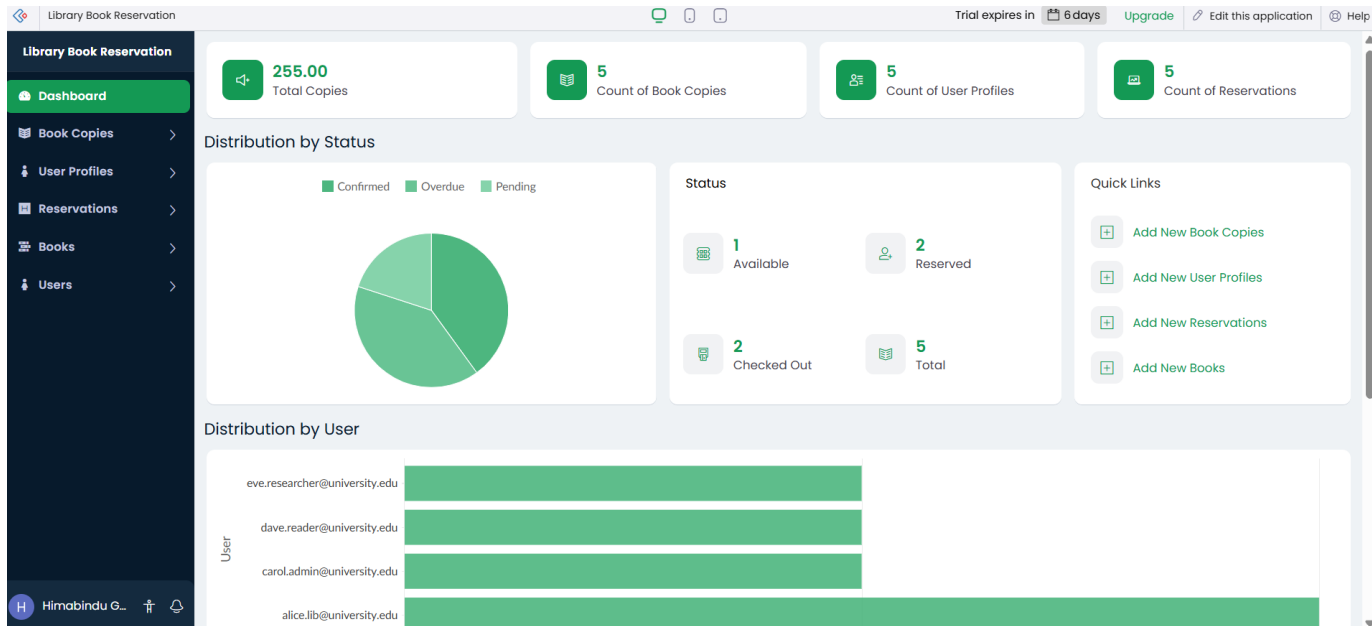
	Book Copy	User	Reservation Date	Due Date	Return Date	Status	Created Time	Last Modified Time
	2 3 4 5	eve345	19-Jul-2026	26-Jul-2026	26-Jul-2026	Confirmed	18-Jul-2026 14:50:00	18-Jul-2026 14:50:00
	1 2 3 4 5	diana012	18-Jul-2026	25-Jul-2026	25-Jul-2026	Confirmed	17-Jul-2026 08:20:00	17-Jul-2026 08:20:00
...	1 2	charlie789	17-Jul-2026	24-Jul-2026	24-Jul-2026	Overdue	16-Jul-2026 11:45:00	16-Jul-2026 11:45:00
	2 3 4	bob456	16-Jul-2026	23-Jul-2026	23-Jul-2026	Pending	15-Jul-2026 10:15:00	15-Jul-2026 10:15:00
	1 2	alice123	15-Jul-2026	22-Jul-2026	22-Jul-2026	Overdue	14-Jul-2026 09:30:00	14-Jul-2026 09:30:00

Showing 5 of 5

Books Report

Book ID

RESULT: The software is been created



Thus, the SIMATS Library Book Reservation System was successfully designed as a cloud-based SaaS application. The system can store and manage book details, check availability, reserve books, and update the status of books such as Taken, Returned, and Available through a web-based interface.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

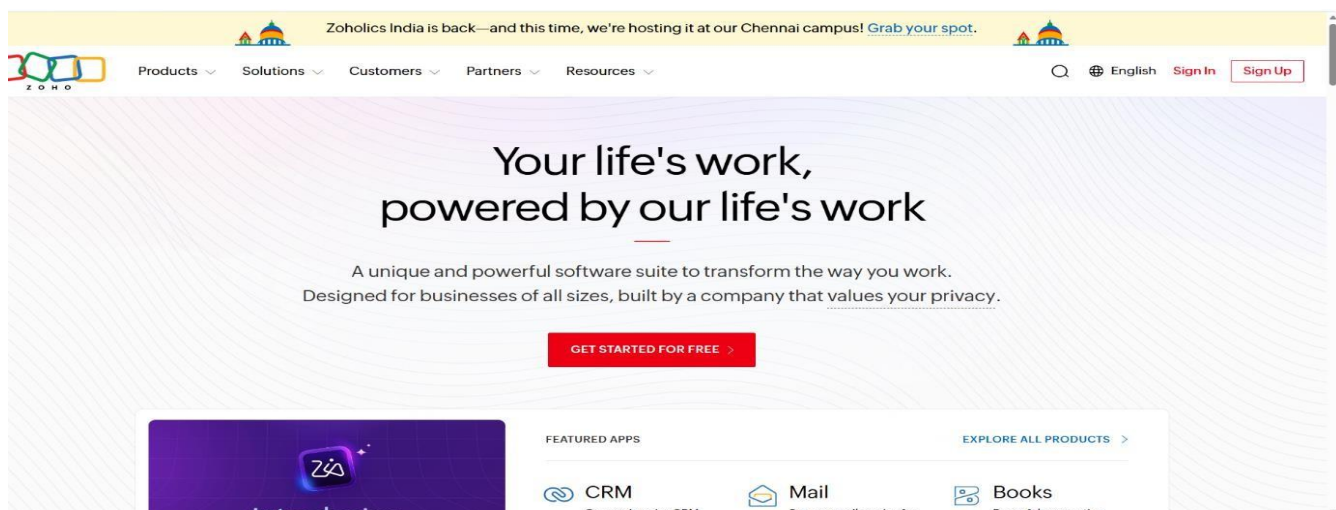
To develop a simple cloud-based Payroll Processing System using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model by managing employee details, salary components, and calculating gross and net salary.

ALGORITHM:

1. Start
2. Open the Payroll Processing System.
3. Enter employee details such as Name, Employee ID, Present Organization Experience, and Overall Experience.
4. Enter salary details such as Basic Pay, HRA, DA, and TA.
5. Calculate Gross Pay by adding Basic Pay, HRA, DA, and TA.
6. Calculate deductions, if applicable.
7. Calculate Net Pay after subtracting deductions from Gross Pay.
8. Store the employee payroll details in the cloud database.
9. Display the employee salary details and Net Pay.
10. Stop.

Implemented:

Step1:Go to Zoho.com



Step2: Login to your zoho account and open your zoho account

 Desk Helpdesk software to deliver great customer support. TRY NOW >	 Payroll Effortless payroll processing software for businesses. TRY NOW >	 Recruit Intuitive recruiting platform built to provide hiring solutions. TRY NOW >	 Projects Manage, track, and collaborate on projects with teams. TRY NOW >
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Apps you've accessed



Cliq



Creator



Forms



People

Step3: Click Create New Application and select Create from Scratch

Create New Application

**Create using Zia**
Build a functional application effortlessly using GenAI
[Create](#)

**Create from scratch**
Create a new application for your business needs
[Create](#)

**Create from gallery**
Pick a pre-built application that suits your requirement
[Pick](#)

**Import from file**
Import your existing data to create an application
[Import](#)

Step4: Enter payroll processing system as a application name and click on create application

**Create from scratch**

Application Name *

☐ **Enable Environment** [Learn More](#)

[Create Application](#)

Step5: :Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

Payroll Processing System

Salary History

Dashboard

Payroll Records

Salary History

Salary History Rep...

Employees

Allowance Config...

Pay Components

Users

Employee

Old Basic Pay

New Basic Pay

Change Date

Reason

Approved By

Submit

Reset

Payroll Processing System

Employees

Dashboard

Payroll Records

Salary History

Employees

Employees Report

Allowance Config...

Pay Components

Users

Name

Date of Joining

Experience in Present Org

Overall Experience

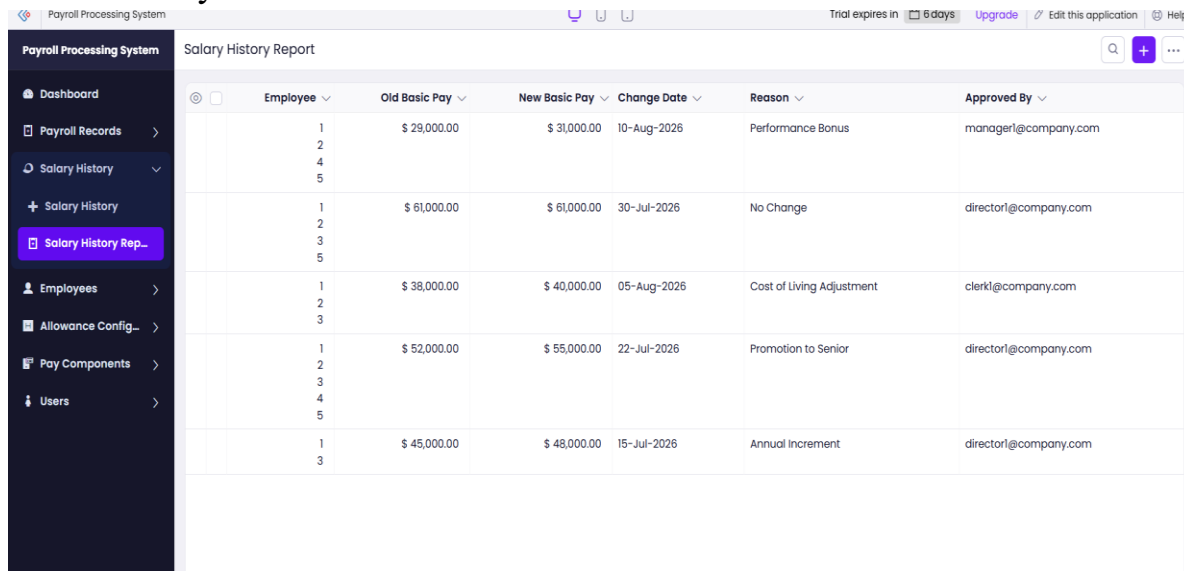
Basic Pay

User

Submit

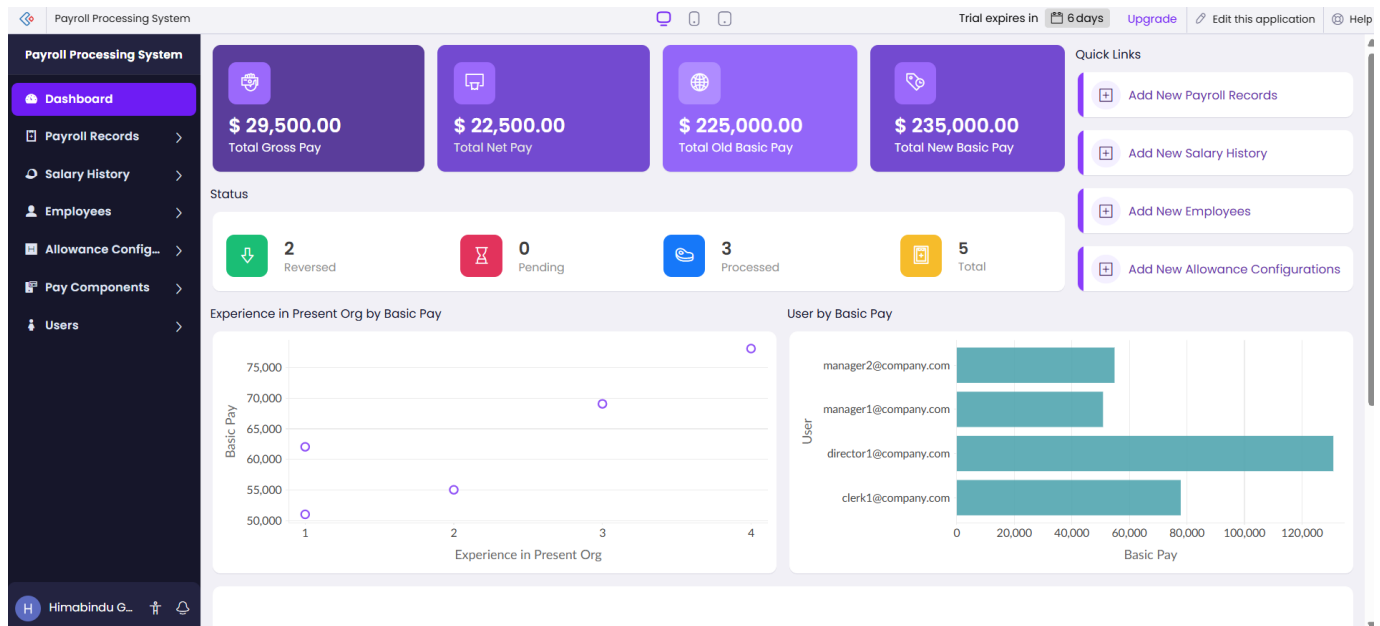
Reset

Step6: Save the form payroll processing ,enter sample booking records and view in the system



Employee	Old Basic Pay	New Basic Pay	Change Date	Reason	Approved By
1 2 4 5	\$ 29,000.00	\$ 31,000.00	10-Aug-2026	Performance Bonus	manager1@company.com
1 2 3 5	\$ 61,000.00	\$ 61,000.00	30-Jul-2026	No Change	director1@company.com
1 2 3	\$ 38,000.00	\$ 40,000.00	05-Aug-2026	Cost of Living Adjustment	clerk1@company.com
1 2 3 4 5	\$ 52,000.00	\$ 55,000.00	22-Jul-2026	Promotion to Senior	director1@company.com
1 3	\$ 45,000.00	\$ 48,000.00	15-Jul-2026	Annual Increment	director1@company.com

RESULT: The Software has been created



Thus, the cloud-based Payroll Processing System was successfully designed as a SaaS application. The system stores employee information in the cloud and calculates HRA, DA, TA, Gross Pay, and Net Pay efficiently. The payroll details can be accessed and managed through a web-based application from different device.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

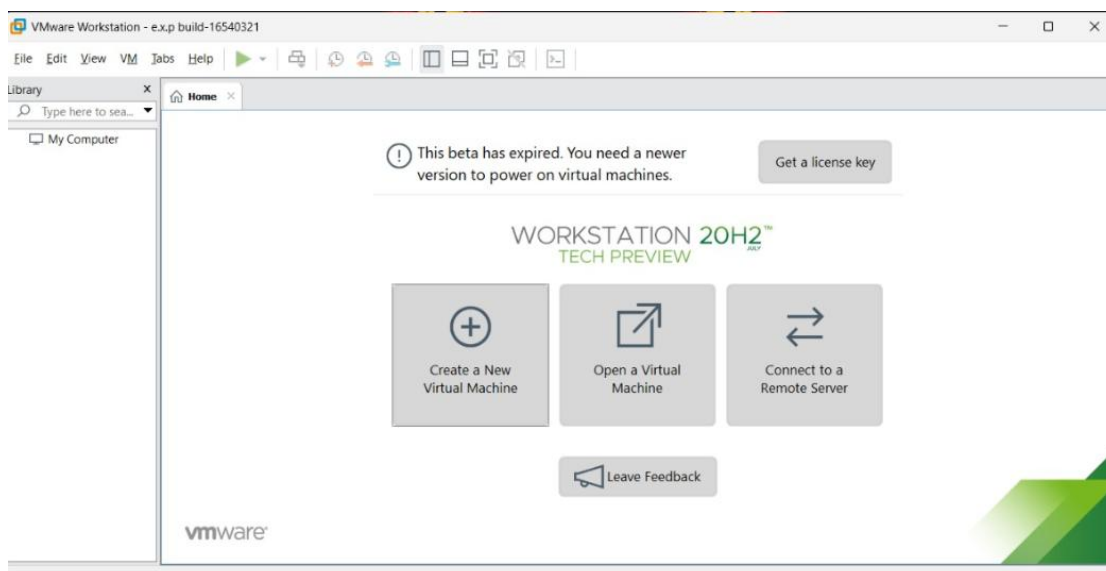
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage.

Steps:

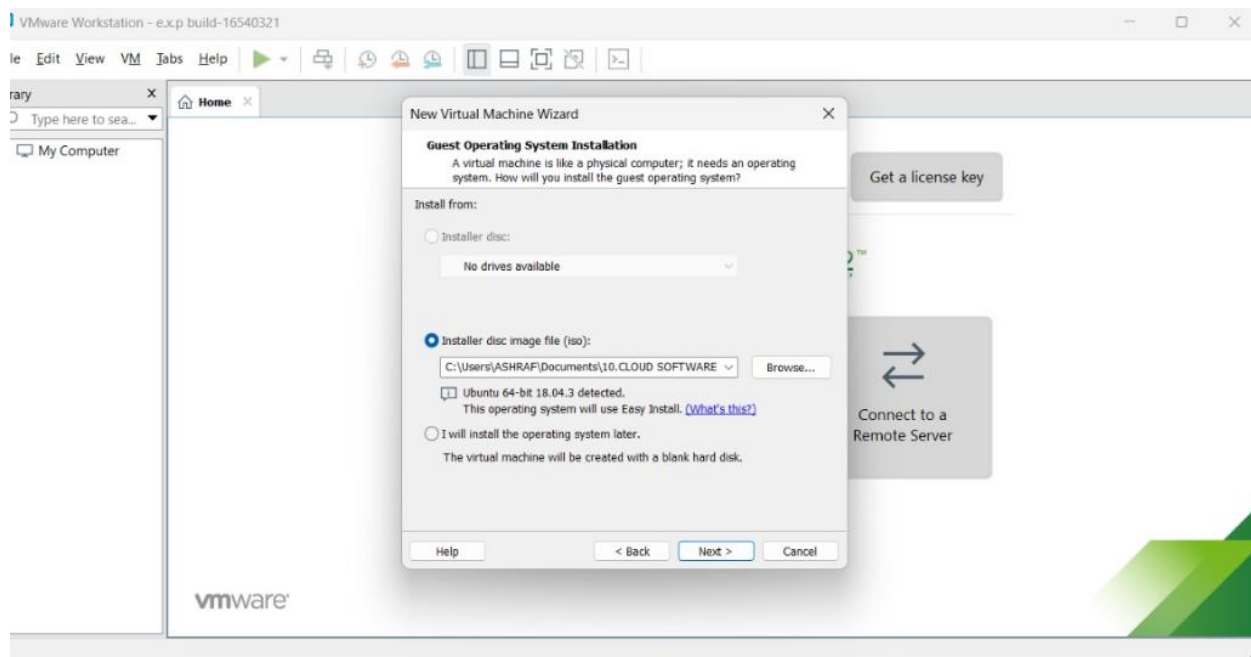
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

Implemented:

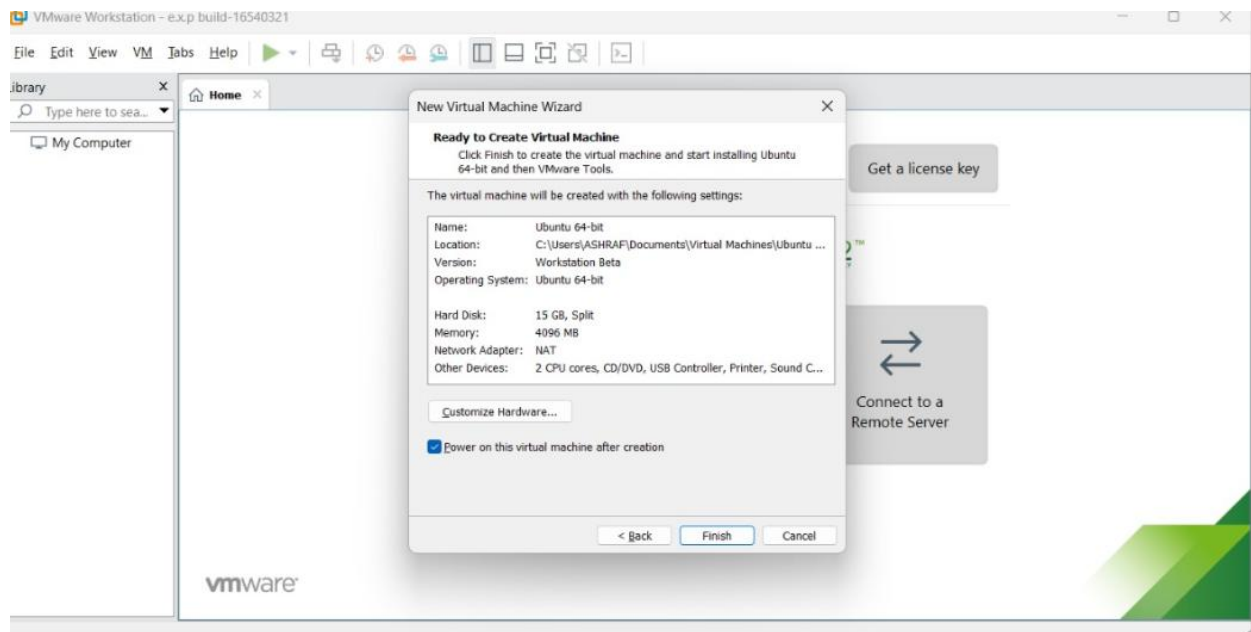
1. Open VMware and select Create New Virtual Machine.



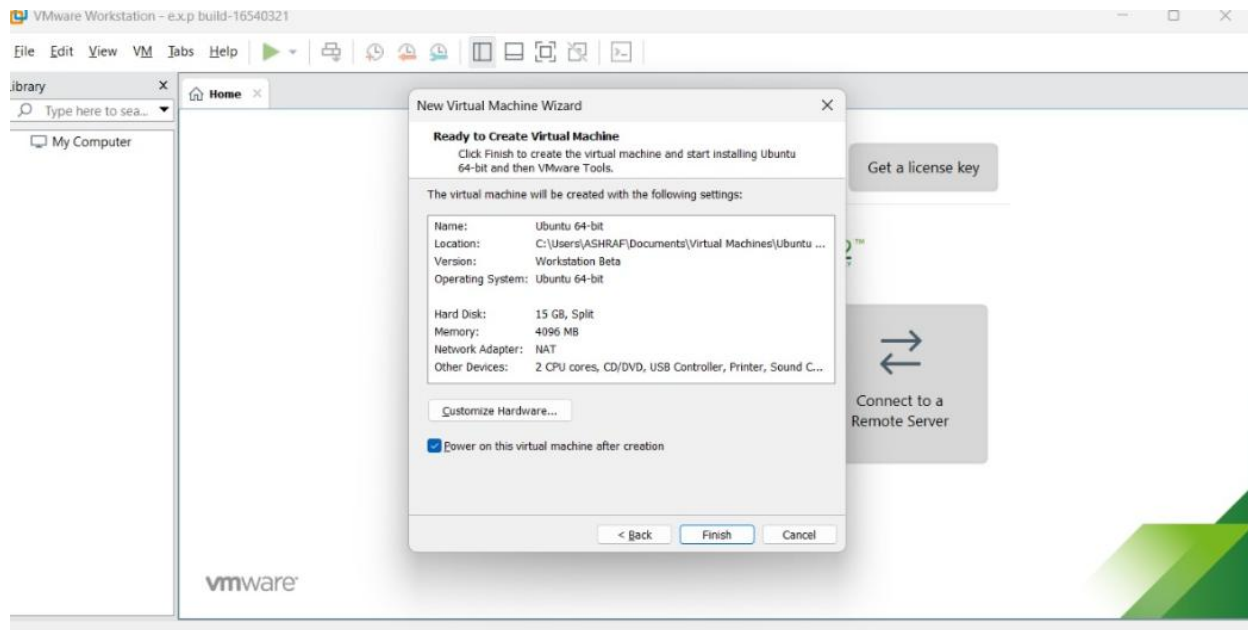
2. Select the Ubuntu ISO file.



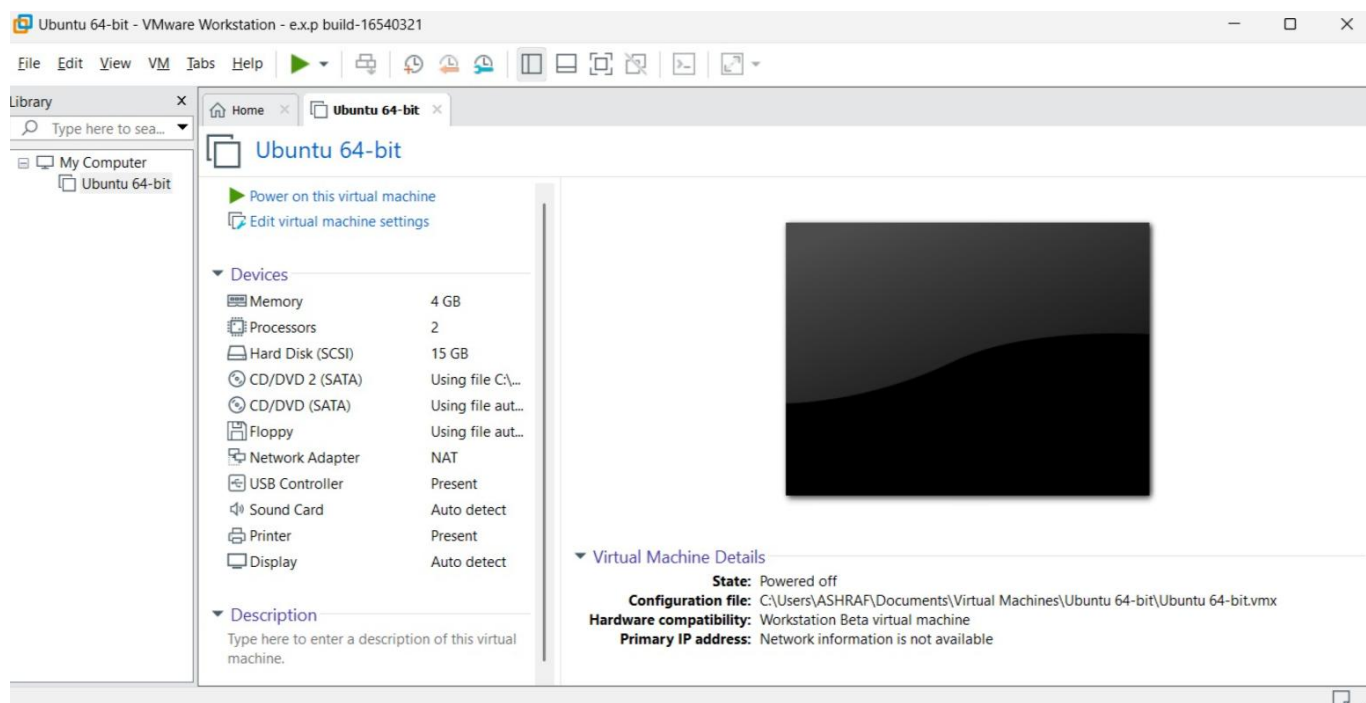
3. Set the VM name as Ubuntu-VM.



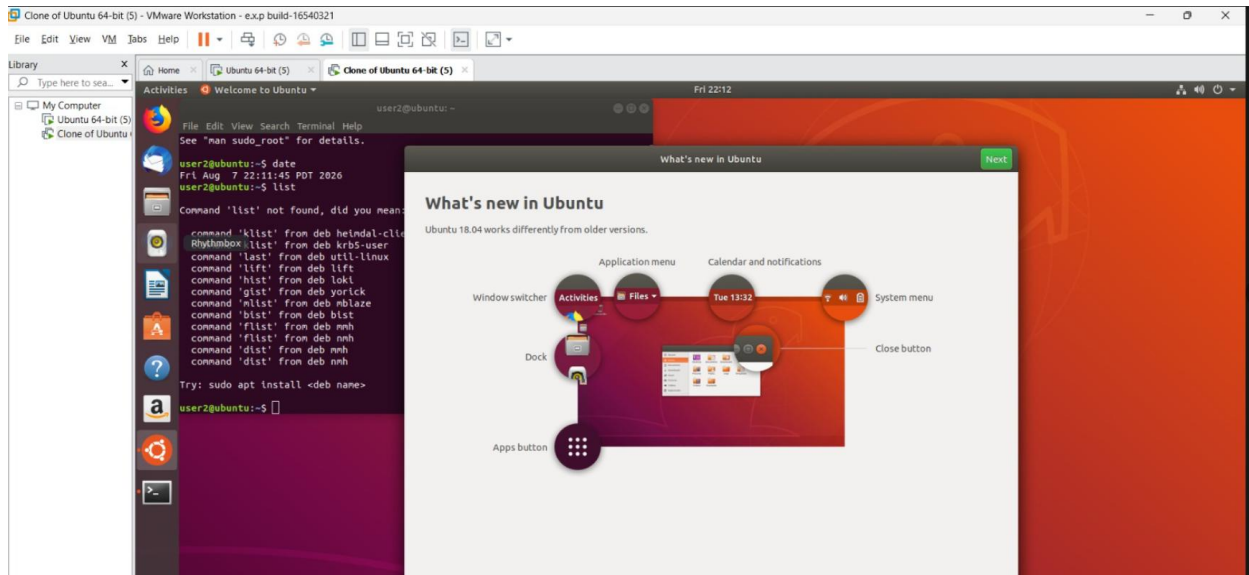
4.Set RAM = 2 GB and CPU = 1.



.5 Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

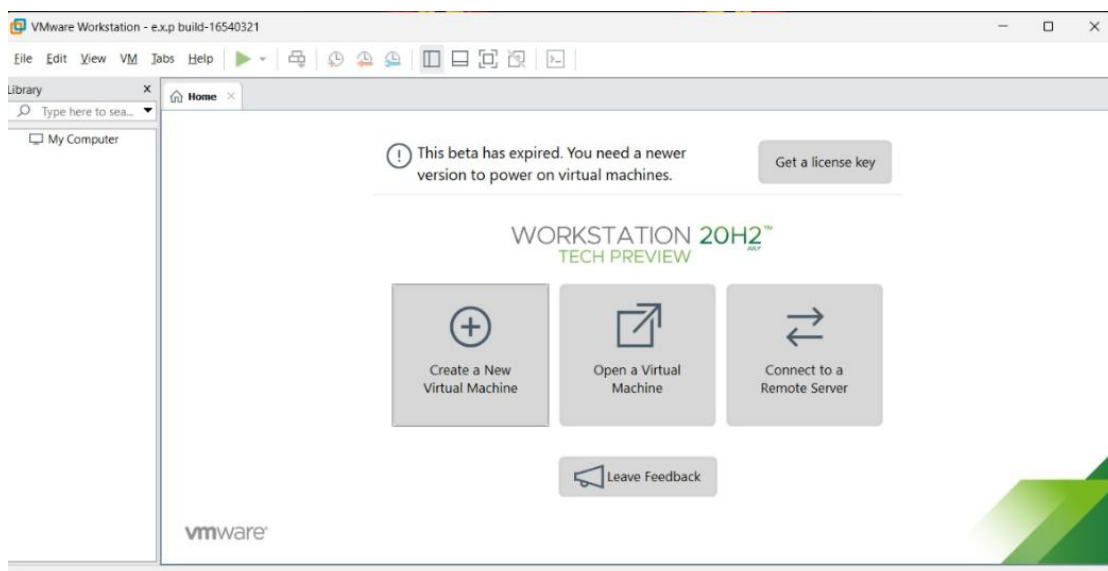
Aim

To demonstrate **virtualization** by installing a **Type-2 Hypervisor (VMware Workstation)**, creating a Virtual Machine with a Windows/Linux operating system, and **cloning the VM** to create and test a previous version of the Virtual Machine.

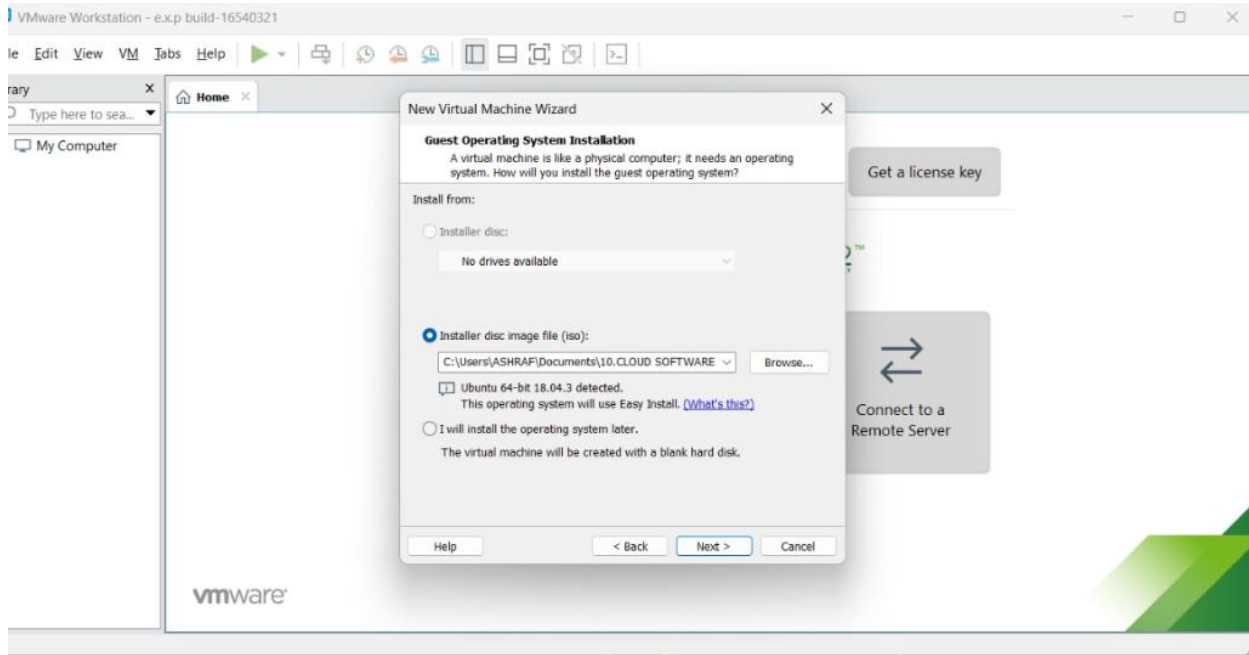
Steps

1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

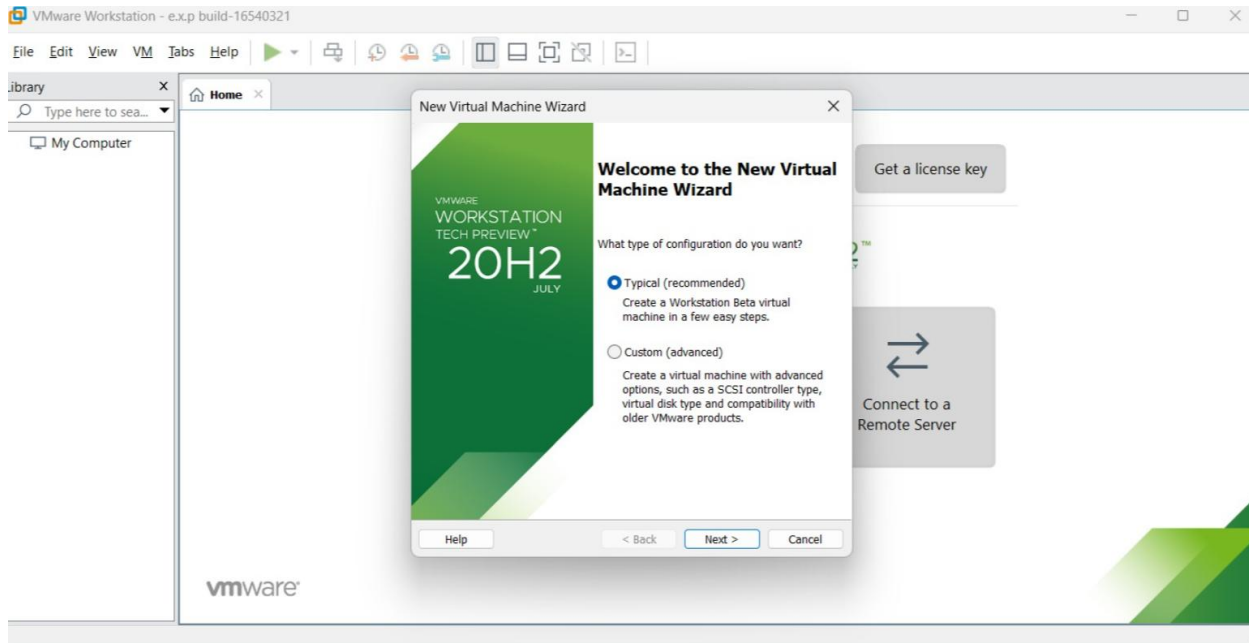
1 Open VMware and select Create New Virtual Machine.



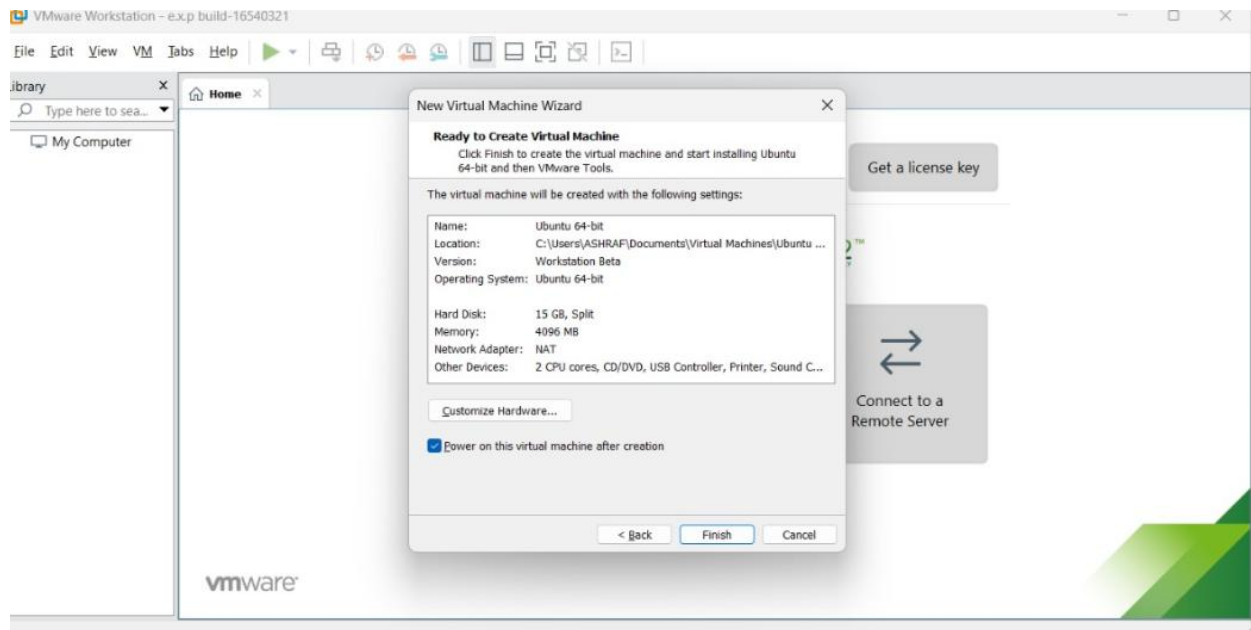
2. Select the Ubuntu ISO file.



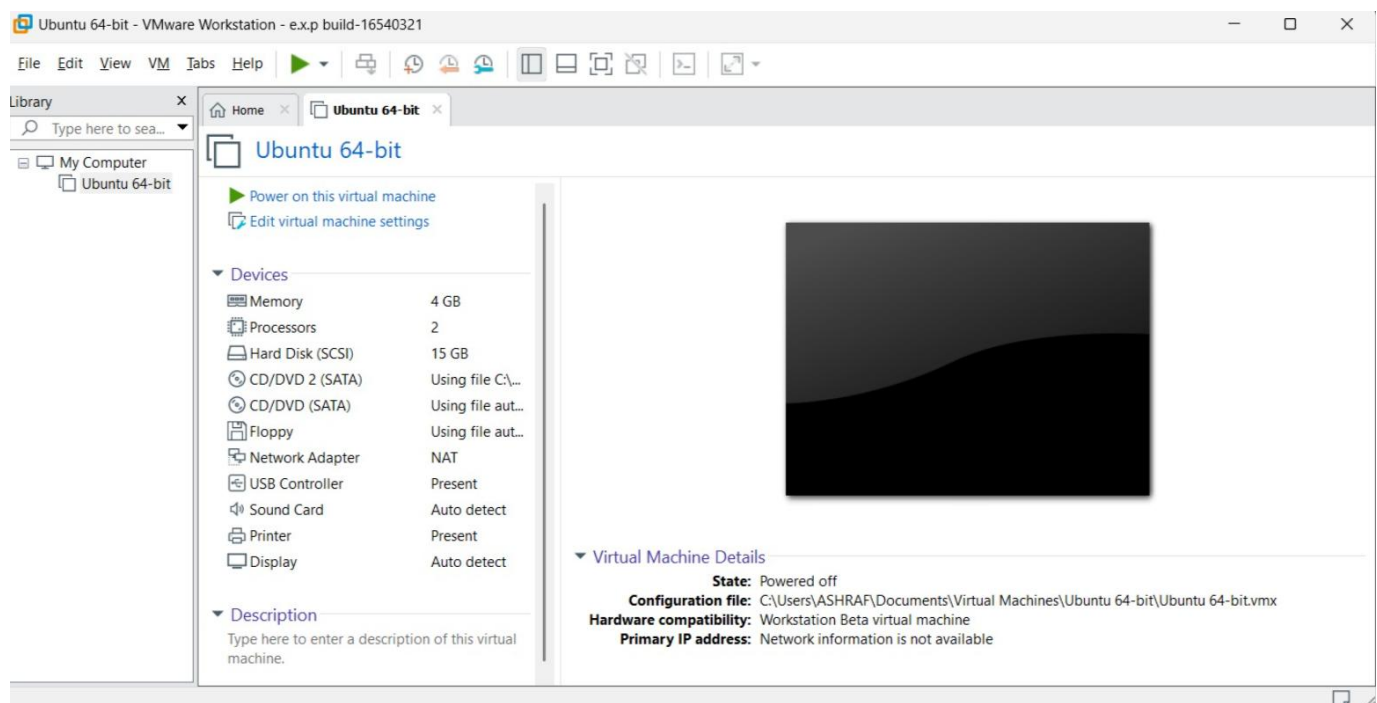
3. Set the VM name as Ubuntu-VM.



4.Set RAM = 2 GB and CPU = 1.



.5 Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.

